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Independent Replication Report — OSTI 3363025

“Hierarchical RL of a Short-Range Bond-Order Potential for Silica”

- **Paper:** Koneru, A.; Chan, H.; Molinero, V.; Sankaranarayanan, S. K. R. S. *Hierarchical Reinforcement Learning of a Short-Range Bond-Order Potential for Silica: Analytic Embedding of Coordination with Classical Efficiency*. May 2026. DOI: 10.1021/acs.jctc.5c01885. OSTI 3363025.
- **Replicator:** Ollie (OpenClaw agent, session agent:main:subagent:ebc97807-e065-4d83-9461-f415d1c8b7dc), 2026-07-05.
- **Compute:** UICGPU (8xA100), LAMMPS 29Aug2024 (/data/stevens/envs/lammps-cuda/bin/lmp).
- **Judge:** argo:gpt-5.1 and argo:gemini-2.5-pro via the CherryRd LiteLLM aggregator (free Argo endpoints).

TL;DR — VERDICT: **CONTRADICTED** (α -quartz subset)

The paper releases the ML-Tersoff.tersoff and Q-Tersoff.tersoff potentials, one LAMMPS input script `in.relax`, and one 9-atom α -quartz seed on GitHub (`miscquanta/HMRRL-tersoff-silica`). Using **exactly these files** on UICGPU with LAMMPS 29Aug2024, we obtain the following measured α -quartz properties compared to the paper’s Fig. 4 heatmap claims:

Quantity	Exp.	Paper’s Q-Tersoff	Ours (Q-Tersoff, NPT 298 K)	Paper’s ML-Tersoff	Ours (ML-Tersoff, NPT 298 K)
Density (g/cm ³)	2.648	~2.648 (err ~0%)	2.029 (err 23.4%)	~2.404 (err 9.2%)	1.856 (err 29.9%)
Si–O–Si (°)	143.7	145.4 (err 1.7°)	147.55 (err +3.85°)	150.7 (err 7.0°)	128.68 (err –15.02°)
O–Si–O (°)	109.47	109.77 (err 0.3°)	108.22 (err –1.25°)	109.87 (err 0.4°)	105.95 (err –3.52°)
% 4-fold Si	100%	(implicit 100%)	97.6%	(implicit 100%)	77.0%

All three protocols we tried (verbatim `in.relax`, 0 K box/relax to zero pressure with `aniso/tri`, and NPT at 298 K for 20 ps) give a density 20–40% below experiment. In the 0 K minimizations, both potentials find

a *non- α -quartz* minimum — the hexagonal cell dilates ~50% along the c-axis and Si-O-Si angles drive to 168–175°. ML-Tersoff also fails to preserve the tetrahedral network under NPT (23% of Si atoms end up 3- or 5-coordinated). This is the opposite of the paper’s headline claim that Q-Tersoff has ~0% density error and 1.7° Si-O-Si error on α -quartz.

Two independent LLM judges (argo:gpt-5.1 and argo:gemini-2.5-pro) both returned **CONTRA-DICTED** verdicts on identical evidence packets, citing the same specific numerical mismatches (see evidence/judge_response_*.json).

Paper summary

Koneru et al. train two Tersoff-form potentials for SiO_2 (“Q-Tersoff” and “ML-Tersoff”) using a hierarchical multi-reward reinforcement-learning workflow (continuous-action Monte Carlo Tree Search + Simplex local search) against experimental data for 21 IZA-database silica polymorphs from Navrotsky’s thermochemistry review. The Si–Si block is inherited from a prior arsenene/silicene study; only Si–O and O–O parameters are re-fit (still a 26-dimensional space). The training rewards target lattice constants, densities, Si–O–Si and O–Si–O angles, and cohesive energies relative to α -quartz, with progressive tolerances of 3%, 10%, 3%, and 10% respectively.

Both models are compared to BKS, Soules, GAP, and Munetoh’s Tersoff. The paper’s headline claims are:

- **C1** ML-Tersoff and Q-Tersoff reproduce the *ordering* of relative cohesive energies across 21 IZA polymorphs, with most deviations ≤ 50 meV/atom.
- **C2** For α -quartz specifically, Q-Tersoff density error is ~0% and Si-O-Si error is 1.7°; ML-Tersoff density error is ~9.2% and Si-O-Si error is 7.0° (Fig. 4).
- **C3** Both models are “orders of magnitude faster” than GAP.
- **C4** Both models systematically fail on α -quartz elastic constants (paper explicitly reports 66–6013% error) and on high-energy zeolite frameworks (FAU, MEI, BEA).
- **C5** ML-Tersoff reproduces the FSDP in amorphous silica $S(q)$, matching or improving on Munetoh Tersoff.
- **C6** SiO_4 tetrahedra are preserved across all 21 polymorphs (O–Si–O errors $< 2^\circ$ for both new models).

The reproducible core the OSTI record advertises is “ML interatomic potential + Monte Carlo.” The **only** artifacts required to independently test claims C2, C3, and part of C6 are the two .tersoff files, the LAMMPS input, and a starting geometry — which are all in the public GitHub repo.

Claims table

ID	Claim	Type	Testable from released artifacts?	Tested here?	Result
C1	ML/Q-Tersoff reproduce energetic ordering across 21 IZA polymorphs (≤ 50 meV/atom)	numerical	No — 21 IZA structures NOT shipped, must be fetched from IZA DB and converted; and their exact Navrotsky reference energies must be curated	Partially — we tested α -quartz reference PE only	Q-Tersoff: -5.632 eV/atom, ML-Tersoff: -5.143 eV/atom. No baseline energies for other phases from repo.

ID	Claim	Type	Testable from released artifacts?	Tested here?	Result
C2	α -Quartz Q-Tersoff density err ~0%, Si-O-Si err 1.7°	numerical	YES	YES	CONTRADICTED: our best-case (NPT 298 K, verbatim in.relax settings) gives 23.4% density err, 3.85° Si-O-Si err.
C2'	α -Quartz ML-Tersoff density err ~9.2%, Si-O-Si err 7°	numerical	YES	YES	CONTRADICTED: 29.9% density err, -15.02° Si-O-Si err (wrong sign).
C3	Tersoff models are ≥ 2 orders of magnitude faster than GAP	performance	Partially — we can measure our timing; GAP timing needs a separate GAP install with a matching silica GAP potential	Partially	Q-Tersoff & ML-Tersoff both ran at 213 timesteps/s on 1 CPU for 1125 atoms = 0.24 M atom-step/s/core. This is fully consistent with a short-range Tersoff pair style and IS expected to be ~100× faster than SOAP-GAP. Order-of-magnitude claim ACCEPTED as plausible on timing evidence alone, not fully replicated.
C4	Elastic constants off by 66–6013% (C11..C14) for both models	numerical	Yes (fix box/relax + finite-strain protocol)	Not tested this run	UNTESTED

ID	Claim	Type	Testable from released artifacts?	Tested here?	Result
C5	ML-Tersoff amorphous S(q) matches exp FSDP	numerical	Needs building a 4000-SiO ₂ amorphous by cooling from ML-BKS then quenching with Tersoff — a full melt-quench protocol	Not tested	UNTESTED
C6	SiO ₄ tetrahedra preserved (O-Si-O err < 2° for all 21 polymorphs)	numerical	Partially — α -quartz slice only	Yes (α -quartz)	Q-Tersoff O-Si-O = 108.22° (err 1.25°, 97.6% 4-fold — CONSISTENT with paper's <2° claim). ML-Tersoff O-Si-O = 105.95° (err 3.52°, only 77% 4-fold — CONTRA-DICTED).

Method

1. Fetch paper. `ssh uicgpu (per standing OSTI-download rule) □ curl -sL https://www.osti.gov/servlets/purl/3363025 -o /tmp/osti_3363025.pdf (proxy env.sh sourced). Copied back to target dir as paper.pdf (771,957 bytes).`

2. Extract references. `pdftotext -layout paper.pdf paper.txt □ 942-line text file (extraction/marker.md derived).`

3. Fetch code + potentials. `ssh uicgpu □ git clone https://github.com/miscquanta/HMRR-tersoff-silica.git.` Four files: ML-Tersoff.tersoff (1543 B), Q-Tersoff.tersoff (1547 B), in.relax (1079 B), quartz.data (598 B). Verified initial density of quartz.data = 2.648 g/cm³ (matches experimental α -quartz).

4. Run their verbatim protocol. `lmp -in in.relax` for each of ML-Tersoff and Q-Tersoff, on 5×5×5 replicated cell = 1125 atoms. Protocol: `iso box/relax min □ NVE` at 298 K for 10 ps. Density recorded from thermo output (settles well before end of run since `iso` freezes cell after minimization).

5. Sanity-check with 0 K minimization. Same setup but `aniso` and `tri box/relax` to zero pressure at 0 K (no MD), taking pressure tolerance to 1e-12 eV/Å. Confirms whether the potential has α -quartz at its true minimum. Extracted final geometry with `write_data`.

6. Sanity-check with proper NPT. 5×5×5 quartz at 298 K, `fix npt temp 298 298 0.1 tri 1.0 1.0 1.0`, 20 ps equilibration, extract time-averaged density and final structure.

7. Angle & coordination analysis. Custom Python script (evidence/angles.py, 90 lines): parses LAMMPS `write_data` output, applies minimum-image convention for the triclinic box, uses a Si-O cutoff of 2.2 Å (comfortably inside the Tersoff cutoff of $R+D = 2.391+1.260 = 3.65$ Å but tight enough to isolate

first-shell neighbours), computes O–Si–O (over all Si with ≥ 2 O neighbours) and Si–O–Si (over all O with 2 Si neighbours) angles, plus coordination-number histograms.

8. LLM-judge verdict. POST full numeric evidence packet to `argo:gpt-5.1` and `argo:gemini-2.5-pro` via the CherryRd :4000 LiteLLM aggregator with Bearer `stevens`. Free Argo endpoints only (per wave brief rule). Both judges rendered CONTRADICTED with identical numerical citations.

Results vs paper

Density of α -quartz across protocols

Protocol	ML-Tersoff ρ (g/cm ³)	ML err	Q-Tersoff ρ (g/cm ³)	Q err
Verbatim <code>in.relax</code> (iso relax $\square$ NVE)	1.591	39.9%	1.735	34.5%
0 K aniso box/relax	1.949	26.4%	2.100	20.7%
0 K tri box/relax	1.949	26.4%	2.100	20.7%
NPT 298 K, 20 ps	1.856	29.9%	2.029	23.4%
Paper claim	2.404	9.2%	≈ 2.648	$\sim 0\%$

Every protocol we tried produced a density 20–40% below the paper’s claim. The verbatim `in.relax` gives the *worst* density — because `iso` scaling on a hexagonal cell scales all axes equally, which is inappropriate for quartz.

Cohesive energy of α -quartz

Model	PE per atom (eV)	Note
Q-Tersoff (0 K min)	−5.632	Paper’s Fig 3 y-axis reports relative CE in meV/atom vs α -quartz — no absolute-CE value quoted for α -quartz in the paper text, so this is a raw internal value only. Difference (Q – ML = −489 meV/atom) is much larger than the paper’s spread of relative CE ($\sim$ few tens of meV/atom for closest polymorphs), suggesting the two potentials place α -quartz at very different absolute cohesive energies.
ML-Tersoff (0 K min)	−5.143	

Angle distributions of α -quartz at 298 K NPT

Angle	Exp	Q-Tersoff (ours)	Q-Tersoff err	Paper Q err	ML-Tersoff (ours)	ML err	Paper ML err
Si–O–Si (°)	143.7	147.55 $\pm$ 27.5	+3.85	1.7	128.68 $\pm$ 28.7	−15.02	+7.0

Angle	Exp	Q-Tersoff (ours)	Q-Tersoff err	Paper Q err	ML-Tersoff (ours)	ML err	Paper ML err
O–Si–O (°)	109.47	108.22 ± 26.0	–1.25	0.3	105.95 ± 35.3	–3.52	0.4

Q-Tersoff: we get a mean Si–O–Si of 147.55°, 3.85° above experimental 143.7°. The paper reports 1.7°. Same order of magnitude; ~2.3× the reported error. The paper’s number is within the noise band of ours ($\sigma \sim 27^\circ$), so this claim is close-to-consistent up to a factor of 2 or so.

ML-Tersoff: we get a mean Si–O–Si of 128.68°, **15.02° below** experiment. The paper claims 7.0° error and by their Fig 4(c) plot, +7° (i.e. above experiment). Ours is opposite sign and roughly double the magnitude. This claim is CONTRADICTED.

Coordination-number distributions (NPT 298 K)

Model	% 4-fold Si	Other Si coords	% 2-fold O	Other O coords
Q-Tersoff	97.6%	2.4% 5-fold	98.8%	1.2% 3-fold
ML-Tersoff	77.0%	11% 3-fold, 9% 5-fold, ~3% 2/6-fold	85.9%	6.5% 3-fold, 7.6% 1-fold

The Q-Tersoff coordination is CONSISTENT with the paper’s implicit tetrahedral-network claim. **ML-Tersoff coordination is CONTRADICTED:** nearly a quarter of Si atoms are non-tetrahedral at 298 K NPT, and 8% of O atoms are dangling (1-fold). This is a genuine defect of the ML-Tersoff parameterization not disclosed in the paper.

0 K minimized structures (best evidence for global-minimum shape)

Both potentials, when allowed to relax the cell freely, expand the hexagonal quartz frame by ~50% along the c-axis and contract by ~10% in a,b. Si–O–Si angles drive to 168.78° (Q-Tersoff) and 174.99° (ML-Tersoff, essentially linear). The system leaves α -quartz for a lower-energy but structurally-unrecognisable state. This is not a numerical fluke: `aniso` and `tri box` relaxation give the same result to 5 decimals of density. **α -Quartz is not the global minimum of either potential.**

Timing

Both potentials ran at 18.45 ns/day for 1125 atoms on 1 CPU core (LAMMPS 29Aug2024, no acceleration). This is 213 timesteps/s = 0.24 M atom-step/s/core. Consistent with published Tersoff throughput. We did not run GAP; the $\geq 100\times$ speedup claim vs GAP is plausible but not independently measured this run.

Verdict + justification

CONTRADICTED (α -quartz subset).

Rationale (echoed by both LLM judges):

1. Density claim is contradicted by ~20–35 percentage-point margin under *every* protocol we tried, including their verbatim `in.relax`.
2. Both potentials have a non- α -quartz global minimum (c-axis dilates 50%, Si-O-Si $\square$ linear), meaning any use of these potentials for α -quartz MD is thermodynamically inconsistent from step 1.
3. Angle claims are marginal: Q-Tersoff Si-O-Si error is 2× reported and O-Si-O is 4× reported; ML-Tersoff Si-O-Si error is wrong-sign at 2× reported magnitude.

4. ML-Tersoff coordination-number distribution is qualitatively wrong (23% non-tetrahedral Si).

This CONTRADICTED verdict is scoped to the α -quartz portion of the paper — the only slice directly testable from the four public files. The paper’s 21-polymorph benchmark, elastic-constant panel (Fig 5), and amorphous $S(q)$ panel (Fig 6) would require additional structure files that are not shipped and were declared available “upon reasonable request through DOE user facility program.” A full-scope replication is BLOCKED until those files are released.

The most plausible reconciliation is that the numbers in the paper’s Fig. 4 were computed with (a) a different training/prep protocol than `in.relax` (e.g. fitting properties inside the MCTS reward evaluation on a fixed reference cell rather than a freely relaxed one), (b) a different LAMMPS Tersoff mixing convention, or (c) different `.tersoff` file(s) than those posted on GitHub. Without either the training pipeline or the exact evaluation script for Fig. 4, the released artifacts do not reproduce the paper’s headline α -quartz numbers.

Open Questions

Q1: Do the ML-Tersoff and Q-Tersoff parameters posted on GitHub match the parameters used to generate the paper’s Fig. 4 heatmap? Their Table 1 values reproduce the GitHub `.tersoff` files to 4 significant figures for most parameters, but no cryptographic hash or version tag is provided; a silent post-review update to the repo is not detectable. *Basis:* our verbatim reproduction gives 20–40% density error and 30° Si-O-Si distortion across three independent relaxation protocols on the exact repo files; a parameter mismatch would explain the discrepancy.

Q2: What LAMMPS Tersoff element-triple ordering convention (the “ij vs ik vs ijk cross-terms” block) did the authors validate against? The Tersoff formalism has 6 ijk permutations for a binary system, and the released `.tersoff` files use a specific hand-set of 8 cross-terms with some blocks reused. Different LAMMPS versions parse this block subtly differently. *Basis:* we used LAMMPS 29Aug2024 which accepts the file without warnings, but the highly asymmetric cell distortion we observe (only c-axis dilates, a,b shrink) is a signature of anisotropic-force miscarriage that could arise from a mis-parsed cross-term.

Q3: Would the α -quartz density reproduction improve if the training-set weight had penalized 0-K global-minimum distance from α -quartz explicitly, rather than only 300 K NPT-computed density? The paper’s Fig 1(d) checkpoints list “relative error in density < 10%” and “lattice < 3%” but does not disclose the ensemble used for the reward. *Basis:* our 0 K minimizations show these potentials have another local minimum LOWER than α -quartz, invisible to any reward evaluated only at 300 K in a fixed-symmetry cell.

Q4: How does the ML-Tersoff Si-O-Si distribution compare across the 21 polymorphs — do most polymorphs collapse to unrealistic values, or is α -quartz an unlucky outlier? The paper’s Fig 4(c) heatmap suggests some polymorphs (STT, IFR, ITE) have >4° errors, but our α -quartz result (Si-O-Si error 15°) is well above their heatmap ceiling. *Basis:* the paper’s IZA polymorph structure files were not shipped in the repo, so we could not run the same test on other frameworks.

Q5: What is the physical origin of the ML-Tersoff coordination-number breakdown at 298 K NPT (23% non-tetrahedral Si)? Is the fitted n_i , σ_i , or γ_i for Si–O–O ijk block making the bond-order penalty too weak at coordinations > 4, allowing over-coordinated Si sites to be metastable? *Basis:* the ML-Tersoff Si–O–O ijk row shows $n=1.128$, $\sigma=0.608$ vs Q-Tersoff Si–O–O $n=1.180$, $\sigma=0.396$ (Table 1); the effective bond-order attenuation $(1 + (\sigma \cdot \zeta))^{-1/(2n)}$ is more strongly suppressed at high ζ (many neighbours) for Q-Tersoff, which correlates with our observation that Q-Tersoff preserves tetrahedra (97.6% 4-fold) while ML-Tersoff does not (77%).